Sobolev Spaces and the Direct Method in the Calculus of Variations

Sarthak Dassarma (template from Professor Hartglass)



Santa Clara University

Table of Contents

1 Classical Variational Formulation

2 Weak Derivatives

3 Sobolev Spaces

4 The Direct Method

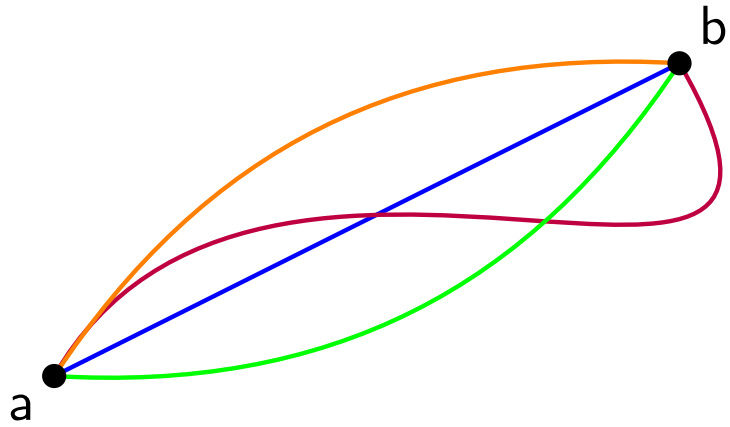
5 Finding the Minimizer

The Shortest Curve Between Two Points

Given two points

$$a, b \in \mathbb{R}^n, \gamma : [0,1] \to \mathbb{R}^n, \gamma(0) = a, \gamma(1) = b$$

what is the shortest curve joining them?



The length of the curve is

$$L(\gamma) = \int_0^1 |\gamma'(t)| dt$$

Question:

$$\min_{\gamma} L(\gamma)?$$

A Problem in the Calculus of Variations

Classically we assume

$$\gamma \in C^1([0,1];\mathbb{R}^n)$$

We wish to minimize the functional

$$L(\gamma) = \int_0^1 |\gamma'(t)|dt$$

Unlike ordinary calculus:

- The unknown is a function
- The space of admissible functions is infinite-dimensional
- We need to understand limits of minimizing sequences

Suppose

$$L(\gamma_k) \to \inf L$$

Does a limit curve exist? If it exists, is it still differentiable?

Table of Contents

1 Classical Variational Formulation

2 Weak Derivatives

3 Sobolev Spaces

4 The Direct Method

5 Finding the Minimizer

Weak Derivatives and the Need for a Larger Function Space

Ideally we want:

- derivatives to make sense
- limits of admissible functions to remain admissible
- compactness for minimizing sequences

Recall the integration by parts formula:

$$\int_\Omega u'(x)\varphi(x)dx = -\int_\Omega u(x)\varphi'(x)dx$$

for all test functions

$$\varphi \in C^\infty(\Omega)$$

The right-hand side still makes sense even when u is not classically differentiable.

Definition of Weak Derivative

Definition

Let $u, v \in L^1_{\text{loc}}(\Omega)$ (if for every compact subset of K $\subset \Omega$, $\int_K |f(x)|dx < \infty$)

We say that v is the weak derivative of u if

$$\int_\Omega u\varphi' = -\int_\Omega v\varphi$$

for every

$$\varphi \in C^\infty(\Omega)$$

We write

$$v = Du$$

The derivative is now defined distributionally rather than pointwise.

Example: A Weak Derivative

Consider

$$u(x) = |x|$$

This function is not differentiable at $x = 0$.
Nevertheless, it has weak derivative

$$Du(x) = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$

Sharp corners are okay, but we *cannot have discontinuous jumps*.

Table of Contents

1 Classical Variational Formulation

2 Weak Derivatives

3 Sobolev Spaces

4 The Direct Method

5 Finding the Minimizer

Sobolev Spaces

Definition

For $1 \leq p \leq \infty$,

$$W^{1,p}(\Omega) = \{u \in L^p(\Omega) : Du \in L^p(\Omega)\}$$

Sobolev spaces enlarge the class of admissible functions while retaining enough differential structure for analysis.

For the remainder of this talk we focus on the case $p = 2$.

Hence

$$W^{1,2}(\Omega) = \left\{u \in L^2(\Omega) : Du \in L^2(\Omega)\right\}$$

The Sobolev Norm

Definition

For $u \in W^{1,2}(\Omega)$, define

$$||u||_{W^{1,2}} = \left(||u||_{L^2}^2 + ||Du||_{L^2}^2\right)^{1/2}$$

This norm measures

- the "size" of the function over $W^{1,2}$
- how fast the function is changing

Definition

A normed vector space $(X, |\cdot|)$ is called a Banach space if every Cauchy sequence in X converges to an element of X.

Fact: $W^{1,p}(\Omega)$ equipped with the Sobolev norm is a Banach space. Even better, $W^{1,2}(\Omega)$ is a Hilbert space, so of course it is complete!

Weak Convergence

Definition

A sequence u_k converges weakly to u in $W^{1,2}(\Omega)$ if $\ell(u_k) \to \ell(u)$ for every continuous linear functional $\ell \in \left(W^{1,2}(\Omega)\right)^*$.

Notation:

$$u_k \rightharpoonup u$$

Compactness

Theorem

Every bounded sequence in $W^{1,2}(\Omega)$ has a weakly convergent subsequence. $|u_k|_{W^{1,2}} \leq C$ implies that some subsequence satisfies $u_{k_j} \rightharpoonup u$.

Closed, bounded sets are generally not compact in the norm topology. However, closed convex sets agree in both the norm and weak toplogies. (This is essential for minimization!)

Table of Contents

1 Classical Variational Formulation

2 Weak Derivatives

3 Sobolev Spaces

4 The Direct Method

5 Finding the Minimizer

Reformulating the Problem

We now consider the energy functional

$$E(\gamma) = \int_0^1 |\gamma'(t)|^2 dt$$

Let

$$\mathcal{A} = \left\{ \gamma \in W^{1,2}([0,1]; \mathbb{R}^n) : \gamma(0) = a, \gamma(1) = b \right\}$$

Our new variational problem is

$$\min_{\gamma \in \mathcal{A}} E(\gamma)$$

Why energy instead of length? Because E behaves much better with respect to weak convergence in $W^{1,2}$.

The Direct Method

The Direct Method consists of three steps:

1. Choose a minimizing sequence

$$E(\gamma_k) \to \inf_{\gamma \in \mathcal{A}} E(\gamma)$$

2. Use compactness to obtain

$$\gamma_{k_j} \rightharpoonup \gamma$$

3. Show that the limit is still a minimizer

Weak compactness gives Step 2.
Weak lower semicontinuity gives Step 3.

Weak Lower Semicontinuity

Theorem

If $\gamma_k \rightharpoonup \gamma$ *in* $W^{1,2}$, *then*

$$E(\gamma) \leq \liminf_{k\to\infty} E(\gamma_k)$$

It ultimately comes from the convexity of

$$v \mapsto |v|^2$$

Therefore energy cannot suddenly decrease in the weak limit.

Existence of a Minimizer

Let (γ_k) be a minimizing sequence:

$$E(\gamma_k) \to \inf_{\mathcal{A}} E$$

The sequence is bounded in $W^{1,2}$.
By weak compactness, after passing to a subsequence,

$$\gamma_k \rightharpoonup \gamma$$

Applying weak lower semicontinuity,

$$E(\gamma) \leq \liminf_{k\to\infty} E(\gamma_k)$$

Since (γ_k) is a minimizing sequence and $\gamma \in \mathcal{A}$, we have:

$$\inf_{\mathcal{A}} E \leq E(\gamma) \leq \lim_{k\to\infty} E(\gamma_k) = \inf_{\mathcal{A}} E$$

Hence, $E(\gamma) = \inf_{\mathcal{A}} E$, and a minimizer exists.

Table of Contents

1 Classical Variational Formulation

2 Weak Derivatives

3 Sobolev Spaces

4 The Direct Method

5 Finding the Minimizer

A Lower Bound for the Energy

Let $\gamma : [0,1] \to \mathbb{R}^n$ be a curve connecting $\gamma(0) = a$ to $\gamma(1) = b$. We define the energy as:

$$E(\gamma) = \int_0^1 |\gamma'(t)|^2 \, dt$$

By the Fundamental Theorem of Calculus, the total displacement is:

$$b - a = \int_0^1 \gamma'(t) \, dt$$

Applying Cauchy–Schwarz:

$$|b-a|^2 = \left| \int_0^1 \gamma'(t) \, dt \right|^2 \leq \left(\int_0^1 |\gamma'(t)|^2 \, dt \right) \left(\int_0^1 1^2 \, dt \right)$$

Since $\int_0^1 1 \, dt = 1$, we obtain the sharp lower bound:

$$|b-a|^2 \leq E(\gamma)$$

The Straight Line Attains the Bound

Consider the straight-line path connecting a and b:

$$\bar{\gamma}(t) = a + t(b - a), \quad t \in [0, 1]$$

Taking the derivative with respect to t, we get:

$$\bar{\gamma}'(t) = b - a$$

Therefore, the energy of this straight line is:

$$E(\bar{\gamma}) = \int_0^1 |\bar{\gamma}'(t)|^2 \, dt = \int_0^1 |b - a|^2 \, dt = |b - a|^2$$

Combining this with our previous lower bound ($|b - a|^2 \leq E(\gamma)$ for all $\gamma \in \mathcal{A}$), we have:

$$E(\bar{\gamma}) = \min_{\gamma \in \mathcal{A}} E(\gamma) = |b - a|^2$$

Thus, the straight line is a minimizer of the energy.

From Energy to Length

By Cauchy–Schwarz, we can relate length to energy:

$$\big(L(\gamma)\big)^2 = \left(\int_0^1 |\gamma'(t)| \cdot 1 \, dt\right)^2 \leq \left(\int_0^1 |\gamma'(t)|^2 \, dt\right)\left(\int_0^1 1^2 \, dt\right) = E(\gamma)$$

For the straight line $\bar{\gamma}(t) = a + t(b-a)$, its length is the Euclidean distance:

$$L(\bar{\gamma}) = \int_0^1 |b-a| \, dt = |b-a|$$

Since we previously showed $|b-a|^2 \leq E(\gamma)$ for all curves, it follows that:

$$|b-a| = L(\bar{\gamma}) \leq L(\gamma) \quad \forall \gamma \in \mathcal{A}$$

Therefore, the straight line is also the shortest curve joining a and b.

Conclusion

Sobolev spaces provide:

- weak derivatives
- completeness
- weak compactness
- existence of minimizers via the Direct Method